

AI Slides Week 1

Chapter 1: Introduction to Artificial Intelligence

Course Overview

- **Instructor:** Noreen Khalid
- **Goals of the Course:**
 - Understand concepts of artificial intelligence (AI) and machine learning (ML).
 - Analyze internal and external environments of AI-based corporations.
 - Explore commercial business models for AI service applications.
 - Identify trends in AI technologies and market demand.
 - Study the full AI course roadmap.

Course Structure Breakdown

- **Unit 1:** The Concept of Artificial Intelligence
 - **Unit 2:** Applications of Artificial Intelligence
 - **Unit 3:** Techniques in Artificial Intelligence
 - **Unit 4:** Artificial Intelligence Trends and Markets
 - **Unit 5:** Course Roadmap
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1.1 Definition of Artificial Intelligence

- **Definition:**
 - AI is the capacity of a machine to perform cognitive functions similar to a human's: thinking and learning.
 - Origin of the term "artificial intelligence" by John McCarthy in 1955.
 - Key processes: learning, reasoning, and self-correction.
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1.2 Types and Subsets of Artificial Intelligence

- **Artificial Narrow Intelligence (ANI):**
 - AI that performs specific tasks (e.g., smart speakers, self-driving cars).
 - Majority of current technology advancements are in this area.
- **Artificial General Intelligence (AGI):**
 - AI that can perform any intellectual task a human can do.
 - Progress in AGI is slower compared to ANI.

Subsets of AI

- **Machine Learning (ML):**

- Subset focused on allowing machines to learn from data and improve autonomously.
 - **Deep Learning:**
 - Subset of ML that leverages multi-layer neural networks for complex data processing.
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1.3 Definition of Machine Learning

- **Definition:**
 - A field within AI that focuses on developing algorithms that let machines learn from data without explicitly being programmed.
 - Notable Definitions:
 - Arthur Samuel (1959): Algorithms that allow machines to learn.
 - Tom Mitchell (1977): Measurement of how well machines perform at specific tasks with increased experience.
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1.4 Disciplines Related to Machine Learning

- **Interdisciplinary Nature:**
 - Combines elements from several fields such as:
 - Probability and statistics
 - Computer science
 - Cognitive science
 - Neuroscience
 - Each relevant discipline enriches the methodologies used in ML.

Comparison with Statistics and Data Mining

- **Machine Learning vs. Statistics:**
 - Statistics: Focuses on theoretical constructs to infer population parameters.
 - Machine Learning: Solves complex tasks with algorithmic patterns rather than strict rules.
 - **Machine Learning vs. Data Mining:**
 - Data Mining: Extracts useful information from large datasets.
 - ML: Engages in learning models for prediction based on data patterns.
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1.5 Types and Choices of Machine Learning-based Data Analysis

- **Data Mining Problems:**
 1. **Prediction:**
 - Models predictions from given data.
 - Example: Predicting quality in fishing industry based on environmental factors.
 2. **Classification:**
 - Classifies data into categories.
 3. **Clustering:**

- Groups similar instances of data.
4. **Association Rule Learning:**
- Identifies relationships between variables in large datasets.

Types of Machine Learning

- **Supervised Learning:**
 - Learning with labeled data. Examples include regression and classification techniques.
 - **Unsupervised Learning:**
 - Learning without labeled outputs, primarily focused on pattern recognition within data.
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1.6 Procedures for Machine Learning-based Data Analysis

1. **Understanding the Business and Defining the Problem:**
 - Clarify the business context and specific problem to solve.
 2. **Collecting Data:**
 - Gather internal and external data relevant for analysis.
 3. **Data Pre-Processing and Searching:**
 - Clean and process data for suitable formats before analysis.
 4. **Data for Model Training:**
 - Train models using suitable algorithms on the prepared data.
 5. **Model Performance Evaluation:**
 - Use separate datasets to validate model performance and adjust parameters.
 6. **Improving Model Performance and Market Application:**
 - Continuous evaluation and enhancement of models.
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1.7 Reasons For Machine Learning

- **Complexity in Traditional Approaches:**
 - Traditional programming struggles with maintaining complex rules.
 - Machine learning simplifies coding and enhances performance.
 - **Adaptability:**
 - ML systems adjust to new data efficiently improving decision-making processes.
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1.8 Limitations of Machine Learning

- **Insufficient Training Data:**
 - High data quantity required for effective ML functionality.
- **Poor Quality Data:**
 - Data with errors negatively impacts learning as patterns become harder to identify.
- **Overfitting & Underfitting:**
 - Issues arise from overly complex models fitting noise in training data or overly simple models failing to capture patterns.

2. Applications of Artificial Intelligence

Example Applications

- **Image Recognition:**
 - Automating identification and categorization of images.
 - **Natural Language Processing (NLP):**
 - Allows computers to understand and respond to human language.
 - **Robotics:**
 - Integrating AI to enhance accuracy and efficiency in industrial automation.
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3. Techniques in Artificial Intelligence

Key Techniques

- **Edge AI:**
 - AI processing on local devices, enhancing speed and reducing latency.
 - **Deep Learning:**
 - Algorithms that learn from large amounts of data through neural networks.
 - **Reinforcement Learning:**
 - Learning through trial and error methods to optimize performance in dynamic contexts.
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4. Artificial Intelligence: Trends and Markets

Future Trends

- **Growth of AI Applications:**
 - Significant potential in sectors such as healthcare, automotive, and finance.
 - **Market Value Expansion:**
 - AI's economic impact predicted to reach \$13 trillion by 2030.
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5. Course Roadmap

- **Key Learning Outcomes:**
 - In-depth understanding of AI concepts and coding for practical application.
 - No focus on reinforcement learning within the limited course timeframe, but encouraged for self-study afterwards.

Category of Machine Learning Techniques

- Various techniques including supervised and unsupervised learning, deep learning algorithms, etc.
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Conclusion

- The course provides foundational concepts, applications, and trends in AI, preparing students for advanced studies in the field.